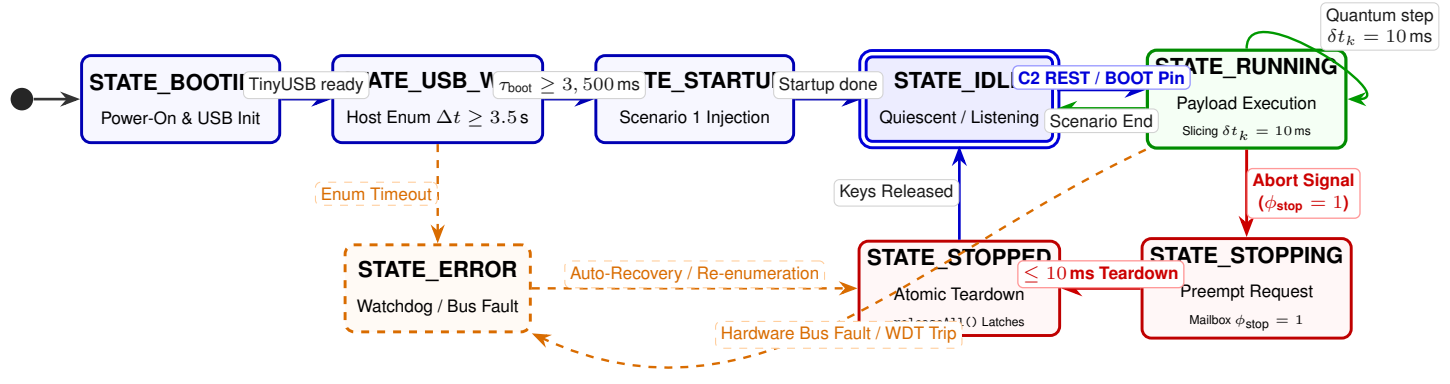

STATE_BOOTII
Power-On & USB Init
TinyUSB ready
ATE_USB_W
Host Enum $\Delta t \geq 3.5\,\mathrm{s}$
$\tau_{\mathrm{boot}} \geq 3,500\,\mathrm{ms}$
TE_STARTU
Scenario 1 Injection
Startup done
STATE_IDL
Quiescent / Listening
C2 REST / BOOT Pin
Scenario End
STATE_RUNNING
Payload Execution
Slicing $\delta t_k = 10\,\mathrm{ms}$
Quantum step
$\delta t_k = 10\,\mathrm{ms}$
Enum Timeout
Keys Released
Abort Signal
($\phi_{\mathrm{stop}} = 1$)
STATE_ERROR
Watchdog / Bus Fault
Auto-Recovery / Re-enumeration
STATE_STOPPED
Atomic Teardown
Latches
$\leq 10\,\mathrm{ms}$ Teardown
STATE_STOPPING
Preempt Request
Mailbox $\phi_{\mathrm{stop}} = 1$
Hardware Bus Fault / WDT Trip